NAME: _____ Geometry (FL B.E.S.T.)

10.3 Density Problems

Density links geometry to real-world materials: $\rho = \frac{m}{V}$ (mass per unit volume), so $m = \rho V$ and $V = \frac{m}{\rho}$ — the same three-way relationship as distance/rate/time. A separate but related idea, **population density**, is people per unit area: $D = \frac{P}{A}$. Most density problems here are two-step: first use a volume formula from this unit to find V (or $B \cdot h$), *then* apply the density relationship. This worksheet also covers comparing densities (which object is denser, or would float in water, since water ≈ 1 g/cm^3), weighted-average density of composite objects, and unit conversions.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Mass 150 g, volume 25 cm^3. Find density.

2. Cube edge 3 cm, density 5 g/cm^3. Find mass.

3. Mass 72 g, density 8 g/cm^3. Find volume.

4. Cylinder radius 2, height 7, density 3 g/cm^3. Find the volume, then the mass (leave π).

5. 12,000 people on 8 km^2. Find the population density.

6. Density 400 people/mi^2, area 15 mi^2. Find the population.

7. Sphere radius 2 cm, density 6 g/cm^3. Find the volume, then the mass (leave π).

8. Object A: 40 g in 8 cm^3. Object B: 45 g in 9 cm^3. Which is denser?

9. Cone radius 3, height 4, density 2 g/cm^3. Find the volume, then the mass (leave π).

10. Population 9000, density 300 people/km^2. Find the area.

11. Mass 200 g, volume 50 cm^3. Find density.

12. Rectangular prism $4 \times 5 \times 6$ cm, mass 360 g. Find its density.

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13. Rearrange $\rho = \dfrac{m}{V}$ to solve for m.

14. A wood block has mass 350 g, volume 500 cm^3. A metal block has mass 800 g, volume 100 cm^3. Find both densities and state which float(s) in water (1 g/cm^3).

15. A statue is cast from bronze (volume 500 cm^3, density 8.9 g/cm^3) fused to a gold-plated cap (volume 20 cm^3, density 19.3 g/cm^3). Find the total mass and the statue's overall density.

✓ Answer Key

1. 6 g/cm^3
2. 135 g
3. 9 cm^3
4. $V = 28\pi$; $m = 84\pi$ g
5. 1500 people/km^2
6. 6000
7. $V = \dfrac{32\pi}{3}$; $m = 64\pi$ g
8. Neither — both are 5 g/cm^3
9. $V = 12\pi$; $m = 24\pi$ g
10. 30 km^2
11. 4 g/cm^3
12. 3 g/cm^3
13. $m = \rho V$
14. Wood: 0.7 g/cm^3, floats; metal: 8 g/cm^3, sinks
15. Total mass 4836 g; overall density 9.3 g/cm^3